

# The Energy Balance Project

*Homeostatic Field Model and Local Actor-Motion Law*

*Foundation Document — Version 2.4 (Foresight vs Artifact; Protective Harvest Rules)*

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**Status:** the V2.3 conclusion is refined by the control it was missing, and three protective harvest rules are added and compared. Backed by scripts and a 24-test suite.

**Central correction.** V2.3 reported that long-horizon foresight destroyed a regenerating resource. That was imprecise: the horizon test there only *gated* a quantity chosen by the immediate ( $H=1$ ) line search. When the quantity is instead chosen to **maximise** the H-tick impact ( $q_{H^*} = \operatorname{argmax}_q I^H(q)$ ), harvesting is fully sustainable (0/32 sources lost, 100% of demand served). So the V2.3 collapse was an **artifact of gating an immediately-optimised quantity**, not evidence that foresight is inherently destructive.

**Practical finding.** A cheap threshold-aware burden term matches the expensive horizon-optimised rule (both sustainable, demand fully served) at  $\sim 1/100$ th the cost, and causes zero regeneration-threshold crossings. And on genuinely randomised clustered layouts, foresight robustly improves routing (+22.3 points over the myopic rule, winning all 20 paired layouts).

## 1. The Control V2.3 Lacked

In V2.3 the regeneration-aware actor (Model D) selected the transfer amount with the immediate burden line search and then used the H-tick counterfactual only to accept or reject it. The acceptance test passes any  $q$  whose predicted impact is positive, including an over-large immediately-optimal  $q$ . The correct, narrower statement of the V2.3 result is therefore:

A long isolated-action ACCEPTANCE test applied to an immediately-optimised harvest quantity can encourage destructive harvesting.

This does not establish that foresight itself is harmful. To decide that, the quantity must itself be chosen by the horizon objective.

## 2. Three Protective Rules and the Decisive Control

### 2.1 Threshold-aware burden (regenerative reserve)

Give each regenerative source a reserve  $R_i$  above its Allee threshold and penalise dropping below it, preserving the convex piecewise-quadratic structure the line search needs:

$$\begin{aligned} R_i &= A_i + \delta_i \\ B_{\text{regen}} &= B + \sum_i \chi_i [R_i - x_i]_+^2 \\ \mu_i^{\text{regen}} &\text{ adds } -2 \chi_i (R_i - x_i) \quad \text{for } x_i < R_i \quad (\text{a depleted source now 'pulls'}) \end{aligned}$$

### 2.2 Hard-reserve baseline

A blunt but clear baseline: cap harvesting at the surplus above the reserve, and forbid it while the source is below its Allee threshold.

$$q_{\text{max}}^{\text{sustainable}} = \max(0, x_i - R_i); \quad \text{harvesting forbidden while } x_i < A_i$$

### 2.3 Horizon-optimised quantity (the control)

Rather than choose  $q$  immediately and ask the horizon yes/no, choose  $q$  to maximise the horizon objective directly:

$$q_{H^*} = \operatorname{argmax}_{\{0 \leq q \leq q_{\text{physical}}\}} I^H(q)$$

If this still over-harvests, the single-action counterfactual is fundamentally biased; if it is sustainable, only the accept/reject architecture was at fault.

## 3. Regeneration Results

Closed Allee economy (8x8, only supply is regeneration,  $A=8 > L=4$ ), 1000 ticks, a 55% source shock at tick 500. Success requires preserving sources AND serving demand. We report both FINAL viability and the mean over the post-shock second half, because the mean alone can hide a late collapse. Sustained recovery (“rec”) counts only if viability returns to  $\geq 90\%$  of its pre-shock value AND source stock stays above the reserve for 100 consecutive ticks.

| Rule                      | viable %<br>(end) | viable %<br>(2nd half) | served % | dead  | A-cross<br>s | stock | rec  |
|---------------------------|-------------------|------------------------|----------|-------|--------------|-------|------|
| safe ( $H=1$ )            | 100.0             | 100.0                  | 100.0    | 0/32  | 1            | 587   | 1t   |
| horizon_gate (V2.3)       | 0.0               | 42.8                   | 73.8     | 32/32 | 60           | 0     | none |
| horizon_opt ( $q_{H^*}$ ) | 100.0             | 100.0                  | 100.0    | 0/32  | 2            | 585   | 5t   |
| threshold_penalty         | 100.0             | 100.0                  | 100.0    | 0/32  | 0            | 587   | 1t   |
| hard_reserve              | 100.0             | 100.0                  | 100.0    | 0/32  | 1            | 587   | 1t   |

|                 |       |       |       |      |   |     |    |
|-----------------|-------|-------|-------|------|---|-----|----|
| penalty_horizon | 100.0 | 100.0 | 100.0 | 0/32 | 0 | 585 | 3t |
|-----------------|-------|-------|-------|------|---|-----|----|

Only horizon\_gate fails, and the end/second-half split makes the failure explicit: its post-shock mean (42.8%) looks survivable, but final viability is 0% and it never sustainably recovers. The horizon-optimised rule is sustainable, proving foresight is not inherently destructive. The two burden-based rules (threshold\_penalty, penalty\_horizon) achieve zero threshold crossings — they never even approach the Allee boundary — and the cheap threshold\_penalty matches the ~100x more expensive horizon\_opt (benchmarked at 0.24 s vs 21.7 s per 200 ticks, ~91x). All sustainable rules serve 100% of demand, so none succeeds by simply refusing to harvest.

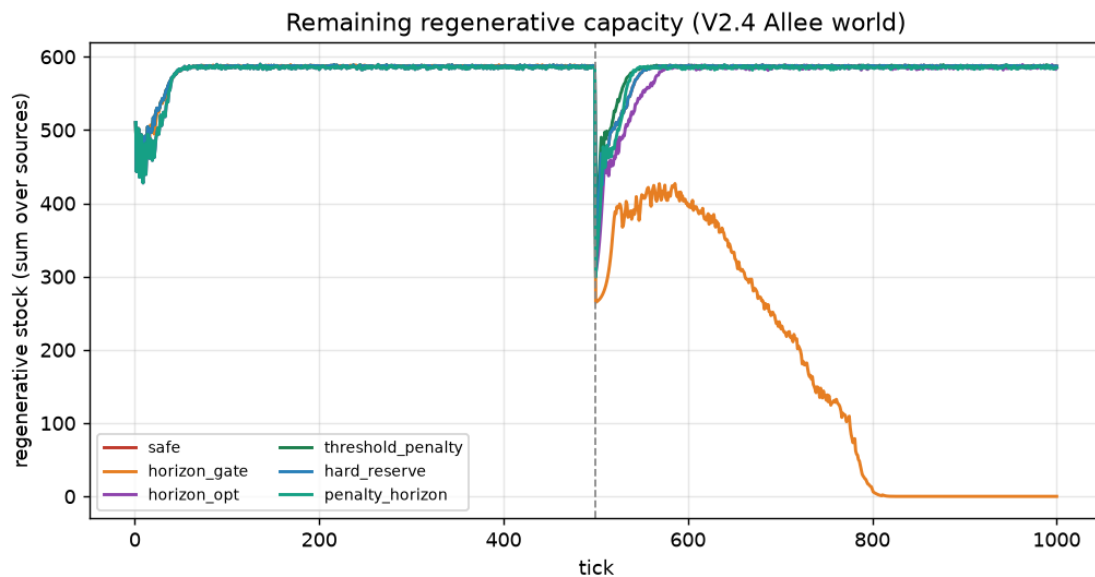


Figure 1. Remaining regenerative stock. After the shock only horizon\_gate (orange) collapses; the horizon-optimised and burden-based rules all recover fully.

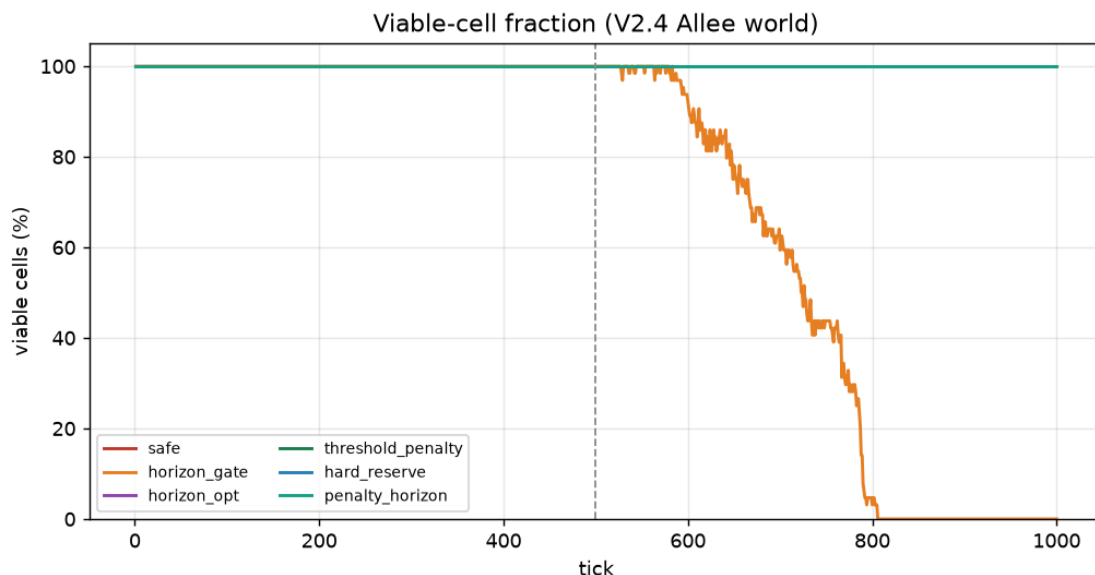


Figure 2. Viable-cell fraction. Five of six rules hold ~100%; only the V2.3 gate collapses.

## 4. Randomised Clustered Audit

The V2.3 clustered layout was deterministic. Here every seed draws a distinct world (random cluster count, centre, radius, density) and every model runs on the identical world for that seed (paired).

Producers are external-flow sources, so the regeneration rules do not apply; we compare transport rules.

| Rule                | viable % (mean $\pm$ sd) | min  | max  | seeds |
|---------------------|--------------------------|------|------|-------|
| none                | 22.0 $\pm$ 11.1          | 9.4  | 54.7 | 20    |
| gradient            | 35.4 $\pm$ 12.8          | 12.1 | 64.0 | 20    |
| safe (H=1)          | 35.5 $\pm$ 13.5          | 14.3 | 63.2 | 20    |
| horizon_gate (H=10) | 57.9 $\pm$ 11.4          | 34.9 | 85.2 | 20    |
| horizon_opt (H=10)  | 58.8 $\pm$ 14.5          | 44.0 | 86.3 | 6     |

The Stage-1 (V2.2.1) finding survives layout randomisation: foresight substantially improves clustered routing. In the paired comparison, horizon\_gate beats the myopic safe rule by +22.3 points on average and wins all 20/20 layouts. (horizon\_opt was run on 6 seeds for cost.)

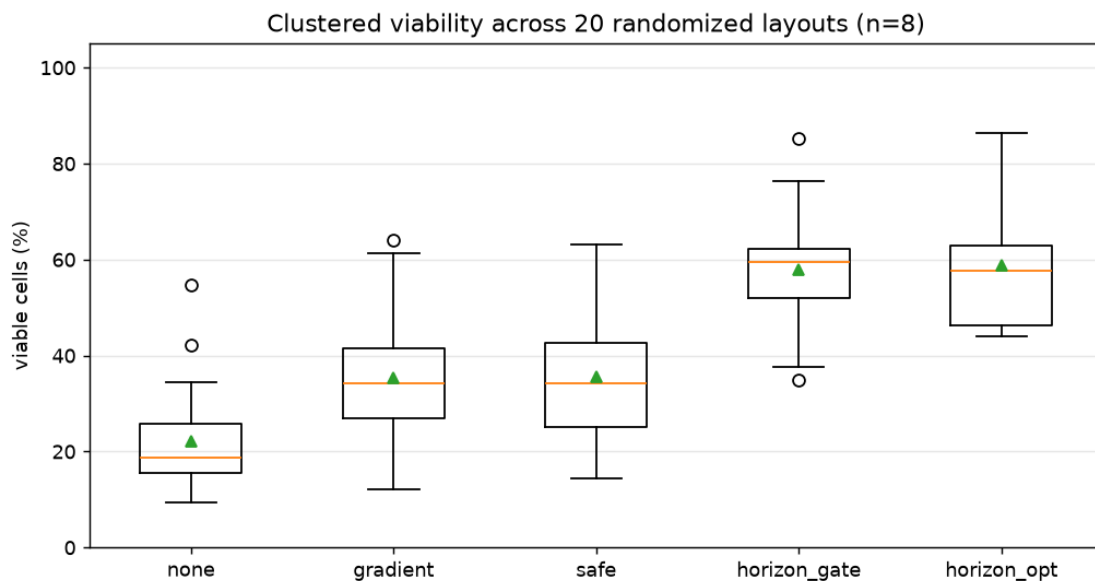


Figure 3. Distribution of clustered viability across 20 randomised paired layouts. Foresight (horizon\_gate / horizon\_opt) clearly shifts the distribution upward.

## 5. Verdict

**Foresight is not the villain; the accept/reject architecture was.** The precise lesson of V2.3–V2.4 is that pairing a long isolated-action acceptance test with an immediately-optimised quantity encourages over-harvest, but the same horizon objective used to CHOOSE the quantity harvests sustainably. Foresight helps for transport routing (clustered) and, correctly formulated, does no harm for resource management.

**The recommended rule is the threshold-aware burden.** It preserves sources and serves all demand, produces zero Allee-threshold crossings, keeps the convex structure the line search needs, and costs ~100x less than the horizon-optimised search. This is exactly the V2.0 Sec. 7.2 design note — a critical regeneration threshold belongs in the burden and regeneration dynamics — now demonstrated rather than assumed. The hard-reserve baseline is equally safe but less flexible; the horizon rules are best reserved for transport routing, where their extra cost buys real benefit.

**Success was defined as preserving sources AND serving demand**, so a do-nothing rule that hoards sources while starving consumers would not have counted; every sustainable rule here serves 100% of demand.

## 6. Limitations and Next Steps

1. The single-action counterfactual still assumes one isolated action; horizon\_opt succeeds here because the objective, maximised over  $q$ , avoids pushing the source below threshold, but in adversarial or strategic settings a mean-field or repeated-action model may be needed.
2. horizon\_opt used a coarse grid + one refinement; a finer optimiser might shift its numbers slightly (not its qualitative sustainability).
3. Clustered horizon\_opt was only 6 seeds for cost; extend for tighter intervals.
4. With a trusted threshold-aware rule in hand, the EBU accounting overlay can finally be added and tested for gaming, hoarding, and repeated credit.

**Reproducibility.** Pure-stdlib core: energy\_balance.py (+ tests). Engines: ebu\_v22.py, ebu\_v23.py, ebu\_v24.py. Experiments: ecosystem.py, experiments\_v22.py, audit\_v231.py, exp\_v23.py, exp\_v24.py, exp\_v24\_clustered.py. Tests (24 total): test\_energy\_balance.py (8), test\_v22.py (7), test\_v23.py (4), test\_v24.py (5). Figures and PDFs are regenerated by the make\_paper\_\*.py scripts. The figure/PDF scripts require matplotlib + reportlab; the core and its physics tests need only the standard library.